Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/01/2026
PI Name:	Greenhalgh, Preston C. M.S.		
Project Title:	Human Volunteer Lunar Landings While in a Standing Posture		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	281-483-6639
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/01/2025	**End Date:**	01/31/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: The end date changed to 01/31/27 per L. Juliette/HFBP (Ed., 6/16/26). NOTE: The start date changed to 07/01/2025 per the PI; original start date was 06/01/2025 (Ed., 8/5/25).		
Key Personnel Changes/Previous PI:	Per the PI, Teresa Reiber was removed from the list of Co-Investigators; Dr. Reiber took a new position in between the submission and the award (Ed., 8/5/25).		
COI Name (Institution):	Newby, Nathaniel (KBR/NASA Johnson Space Center) Yates, Keegan (KBR/NASA Johnson Space Center) Young, Millennia (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This work seeks to replicate lunar landings by dropping human volunteers on a human-rated test device while standing. The data collected will be used to overcome many of the limitations of the current NASA requirements governing crew landings while standing, and address gaps relating to crew in dynamic environments, by investigating human performance and by creating, training, and validating human models. The proposed research will help address these gaps through two aims: 1) quantify the response of human volunteers to vertical loading while standing, and 2) estimate parameters for a standing human spring-mass-damper model. The limitations of current NASA standards addressing standing landings include lack of data and lack of robust analogs within that data. The human drop tests and their various conditions conducted in this study will add a substantial amount of human performance and kinematic data and will provide the most accurate analog possible to replicate lunar landings while standing. While these human volunteer tests will be conducted at non-injurious levels, they will determine loading conditions that are tolerable as related to human performance. Because of the appropriateness of very low injury risk for lunar missions, a better understanding of the human tolerance to these drop conditions will act as a barrier to protect against injury. Each subject will conduct simulated lunar landings with various conditions, including landing pulse, landing angle, and restraint type. Various methods of assessment will be used to qualitatively and quantitatively measure performance of each subject from each simulated landing. Subjects will be outfitted with IMUs (inertial measurement units), and accelerometers and load cells will measure loads and moments of each subject's reaction to simulated lunar landings. This data, combined with survival analysis methodology, will associate variables that accurately predict performance loss. The data collected from the instrumented subjects will also be provided to other funded research studies investigating computational standing landing model verification and validation. This study will provide a robust data set with many test subjects and many test conditions that is needed to train and verify these models. The subject performance analysis, coupled with instrumentation data, will also be used to estimate parameters for a standing human spring-mass-damper model. These types of models are already used in spaceflight to estimate the human response and risk of injury while seated. This work will create a new model capable of estimating the human response while standing and estimate loss of performance. This type of model is simple but provides a straightforward method of prediction that is applicable to a large range of potential landing conditions that are not possible to fully replicate with other complex methods. Lastly, the study will assess the effects of restraints and suits on human performance during standing landings. Comparisons between restrained and unrestrained conditions will be used to measure the effectiveness of basic restraint systems to improve human performance. While the proposed work is not intended to include an exhaustive study of the effects of restraints and suits on crewmember performance during standing landings, the results from this study will inform how future studies could more extensively study the effect of restraints and suits on human performance. Because the findings of this study involve understanding crew performance and injury risk, the findings and impacts from this study are applicable beyond the NASA Human Research Program (HRP). Additional customers include the Human Landing System (HLS) program, HLS providers, the Health and Medical Technical Authority (HMTA), and the Office of the Chief Health and Medical Officer (OCHMO).
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	The research from this study helps ensure the safety of astronauts and other crewmembers taking part in spaceflight to the Moon. Specifically, this research will help better understand human tolerance to dynamic impacts while in a standing posture, to ensure that the mission can be completed ,and to minimize injuries that could cause complications to life back on Earth.
Task Progress:	For the period of this report, this project remains in the pre-testing planning/preparation stage. The main accomplishment for this study was completing all the necessary planning and preparation to submit the necessary protocol paperwork to the Institutional Review Board (IRB), which reviews the ethics and safety for testing humans. To complete the IRB protocol, the test matrix needed to be finalized so that the necessary software, hardware, and test devices could be accounted for and planned for in detail. The test matrix has three main variables: acceleration pulse, restraint method, and landing orientation. The finalization of the test matrix and other planning led to the submission and acceptance of the IRB. Concurrently with the submission of the IRB, other tasks required for the preparation of testing began, including designing and building the drop tower platform, equipment/hardware refinement, creation of test protocols, and subject recruitment.
Bibliography Type:	Description: (Last Updated: 05/20/2026)
Abstracts for Journals and Proceedings	Greenhalgh PC, Yates KM, Newby NJ. "Human volunteer lunar landings while in a standing posture." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026